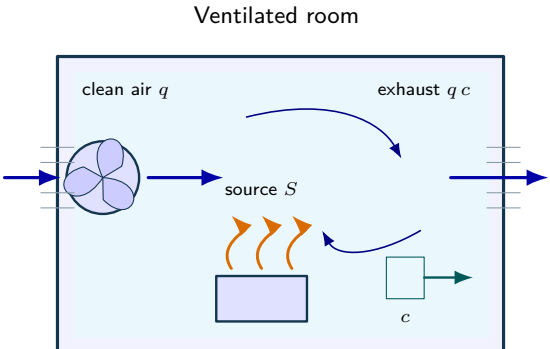

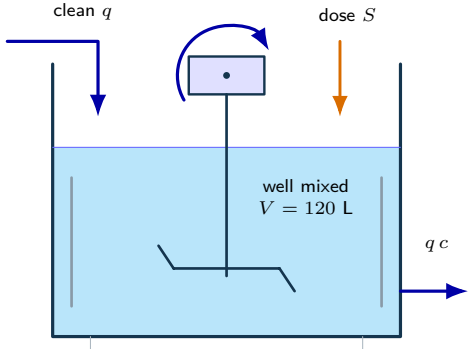

$V = 120$ m^3; concentration c

Liquid analogue: same balance, different units

Inventory and channels

c: tracer concentration.
S: source rate; q: throughput.
Equal inlet/outlet flow fixes V.

Balance and bilinearity

$V\dot{c} = S - qc$.
The product qc couples state and input. Room and liquid cases use different volume units.

Mixing assumption

Grilles and baffles illustrate ideal mixing only. Air velocity fields, fan dynamics and mixing transients are not modeled.